

Notes: Einav, Jenkins, and Levin (Econometrica, 2012)

Contract Pricing in Consumer Credit Markets

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1 Big picture

- **Question.** How should a lender price credit contracts when the contract terms affect not only the number of borrowers, but also borrower selection, leverage, default, and realized revenue?
- **Setting.** The paper studies a large subprime auto lender that sells used cars and finances purchases for borrowers with low income or poor credit histories. The market is attractive for studying credit-contract pricing because default is common, liquidity constraints are severe, and borrower quality is only partially captured by observed credit scores.
- **Core data contribution.** The authors observe detailed transaction-level data on loan applications, contract offers, car characteristics, negotiated prices, loan sizes, down payments, repayment histories, default, recovery, and firm cost. The data include both buyers and nonbuyers, which is crucial for estimating demand.
- **Main economic mechanism.** Down payment requirements work through two channels: they screen out high-risk marginal borrowers and reduce the loan size for infra-marginal borrowers. By contrast, higher car markups are mostly financed, so they increase loan balances and monthly obligations rather than sharply reducing sales.
- **Identification idea.** The paper exploits discrete changes in the lender's down payment schedule and list-price markup formula. These pricing changes identify how purchasing, borrowing, and repayment respond to different contract terms.
- **Main demand result.** A \$100 increase in the minimum down payment lowers the probability of sale by about 2.3 percentage points, while a \$100 increase in car price has little economic effect on sale probability. Consumers are highly sensitive to cash needed today, but much less sensitive to price increases that can be financed.
- **Main repayment result.** A larger loan substantially worsens repayment. The estimates imply that a \$1000 larger loan reduces the fraction of payments made by roughly 32 percent.
- **Contract-design result.** Optimal down payment requirements should be risk based: lower for safer borrowers and higher for riskier borrowers. The lender's observed contracts have this broad feature, but the paper finds that pricing could have used credit-score information more aggressively.
- **Value of information.** Relative to optimal uniform down payments, risk-based financing raises expected profits; with perfect information about borrower private characteristics, profits would rise much more, showing that important private information remains unpriced.
- **Economic takeaway.** In consumer credit markets, contract terms are multidimensional prices. A down payment is not equivalent to a markup or interest-rate change: each term affects demand, selection, leverage, and repayment through different margins.

2 Incremental contribution of the paper

- **From testing asymmetric information to pricing contracts.** Much of the empirical literature asks whether adverse selection or moral hazard exists. This paper goes further by estimating how asymmetric information changes optimal contract terms and lender profitability.
- **A unified model of purchase, borrowing, and repayment.** The paper links three decisions that are often studied separately: whether to buy, how much to borrow, and whether to repay. This is essential because the same contract term can affect both borrower composition and later default incentives.
- **A contract-pricing framework for consumer credit.** The paper adapts standard demand-and-supply tools from industrial organization to a setting in which firm profits depend on who the customers are and how they behave after purchase.
- **Separation of contract dimensions.** The paper shows that down payment requirements and car markups have very different economic roles. Down payments primarily screen and discipline leverage; markups primarily change loan size and repayment burden.
- **Direct quantification of credit scoring.** The counterfactual analysis measures the value of using risk categories to set down payment requirements and also shows how proprietary credit information can create a strategic entry barrier.
- **Policy relevance.** The findings speak directly to consumer-credit booms: relaxing down payment requirements can quickly expand borrowing while worsening the quality and fragility of the borrower pool.

3 Data and measurement

3.1 Subprime auto-lending environment

The firm operates used-car dealerships and finances purchases for high-risk consumers. Customers fill out loan applications, are assigned proprietary credit categories, and receive financing terms. Most customers who buy finance a large share of the car purchase.

- The sample covers loan applications and sales from June 2001 through December 2004.
- The data contain more than 50,000 loan applications; for estimation, the authors use a random subsample of 45,000 applicants to reduce computation time.
- Applicants have low incomes and limited access to standard credit. Many have no bank account, no FICO score, or a score below common prime-lending thresholds.
- Roughly one-third of applicants purchase a car.

Interpretation: This is a setting in which liquidity constraints, adverse selection, and repayment incentives are all economically first order. The lender cannot evaluate profitability from demand alone; it must also forecast default and recovery.

3.2 Contract terms and borrower choices

The relevant contract dimensions include the car price, the minimum down payment, the interest rate, and the loan term.

- The firm sets list prices using a formula based on car cost, but negotiated transaction prices can differ from list prices.
- The minimum down payment depends on the borrower's credit category and changes discretely over time.
- Buyers may pay more than the minimum down payment, but many choose exactly the required minimum.
- The loan amount is approximately the transaction price minus the down payment, plus taxes and fees.

Interpretation: The down payment requirement is both a liquidity screen and a leverage constraint. A consumer who cannot produce more cash today may either not buy or may borrow heavily, which affects subsequent default risk.

3.3 Loan outcomes and revenue measurement

The paper measures repayment outcomes using the fraction of scheduled payments made, default, recovery after repossession, and operating revenue.

- Among uncensored loans, 61 percent end in default.
- Defaults occur early: many happen before a quarter or half of payments have been made.
- Recoveries are low relative to car cost, especially after default.
- Per-sale profitability is highly bimodal: paid loans are profitable, defaulted loans are often unprofitable.

Interpretation: The lender's pricing problem is not simply a markup problem. The central object is expected profit per applicant, which combines close rates, loan size, payment duration, recovery, and cost.

3.4 Risk categories and private information

The firm uses a proprietary credit score to classify borrowers into risk categories. These categories are strongly predictive of repayment, but not sufficient.

- Low-risk borrowers default less often than high-risk borrowers.
- Conditional on observed risk category, borrowers who voluntarily make larger down payments are better risks.
- This implies that borrowers' down payment choices contain private information about repayment risk.

Interpretation: The paper separates observed risk, private information revealed by borrower choices, and the causal effect of leverage on repayment.

4 Models and empirical specifications

4.1 Contract-pricing problem

Let a contract be denoted by ϕ , buyer characteristics by ζ , and the seller's per-sale profit by $\pi(\phi, y(\phi, \zeta))$. Total demand is

$$Q(\phi) = \int 1\{U(\phi, \zeta) \geq U(\zeta)\} dF(\zeta),$$

and expected profit can be written as

$$\Pi(\phi) = Q(\phi)R(\phi),$$

where $R(\phi)$ is expected profit conditional on sale.

For a one-dimensional contract term, the optimality condition is

$$\frac{R(\phi)/\phi}{dR(\phi)/d\phi} = -\frac{Q(\phi)/\phi}{dQ(\phi)/d\phi}.$$

Interpretation: This generalizes the standard monopoly markup condition. Here, changing a contract term affects not only quantity demanded, but also borrower selection and later repayment behavior.

4.2 Consumer borrowing and repayment model

A consumer chooses whether to buy, how much to pay down, and whether to continue repaying. If the car price is p , the minimum down payment is d , and the chosen down payment is D , then loan size is

$$L = p - D.$$

For a loan of size L , the monthly payment is

$$m(L) = \frac{zL}{1 - (1 + z)^{-T}},$$

where z is the monthly interest rate and T is the loan term.

The consumer defaults when the value of keeping the car and making payments falls below the value of default. The model implies that lower liquidity, lower car value, and larger loan size make default more likely.

4.3 Linear estimating equations

The authors approximate the consumer model with three linked estimating equations. The purchase equation is

$$Q = 1\{x'\alpha_x + \alpha_p p + \alpha_d I(D^* \leq d)d + \varepsilon_Q \geq 0\}.$$

The desired down payment is

$$D = \max\{D^* = x'\beta_x + \beta_p p + \varepsilon_D, d\}.$$

The repayment equation is

$$\frac{S}{T} = \min\{\exp(x'\gamma_x + \gamma_L L + \varepsilon_S), 1\}.$$

- α_d measures the sensitivity of purchase to binding down payment requirements.
- β_p measures how much a higher price translates into a larger down payment rather than a larger loan.
- γ_L measures the causal link from loan size to repayment duration.
- Correlations among ε_Q , ε_D , and ε_S capture selection on unobserved borrower quality.

Interpretation: The specification is deliberately close to observables. It estimates the exact elasticities needed for pricing, without requiring full structural estimation of primitive preferences.

4.4 Price negotiation equation

The negotiated price p is related to the list price l by

$$p = x'\lambda_x + \lambda_l l + \varepsilon_p.$$

The list price serves as an instrument for negotiated price because bargaining outcomes may be correlated with unobserved buyer characteristics.

Interpretation: This is important because naive use of negotiated prices could confound price effects with consumer type or dealer information.

4.5 Profit and recovery

Expected profit from a sale is

$$\pi = D + \left(\frac{1}{\kappa} (1 - e^{-\kappa S}) \frac{1}{z} (1 - e^{-zT}) \right) (p - D) + e^{-\kappa S} k(S) - C.$$

The first term is the down payment, the second term is the present value of loan payments, the third term is expected recovery after default, and the final term is cost.

Interpretation: A more generous loan can increase revenues if payments are made, but it also raises default probability and reduces expected realized revenue.

5 Identification and empirical design

5.1 Variation in down payment requirements

The firm changed its down payment schedule more than 20 times during the sample period. These changes differed across credit categories and occurred at discrete points in time.

- The paper includes year, month, city, and credit-grade controls.
- Identification comes from abrupt changes in required down payments, not slow-moving seasonal or demand trends.
- The same variation affects purchase probability, down payment choice, and loan size, helping identify repayment effects.

Interpretation: This strategy is close to a pricing experiment: otherwise similar applicants face different down payment requirements because of policy changes.

5.2 Variation in car markups

The firm changed the list-price markup formula twice. Because the formula maps car cost into list prices, it creates price variation across otherwise comparable cars and over time.

- Negotiated price is treated as potentially endogenous.
- List price changes are used as an instrument for transaction price.
- The paper focuses on price and down payment rather than interest rates because the identifying variation for interest rates and loan terms is weaker.

Interpretation: The distinction between list-price variation and negotiated-price variation strengthens the identification of price effects.

5.3 Selection versus repayment incentives

The key empirical challenge is that borrowers who choose lower down payments may be inherently riskier, but lower down payments also mechanically increase loan size and make default more likely.

- Selection is captured by correlations between unobserved purchase/down-payment components and repayment components.
- The causal effect of leverage is captured by how loan size affects payment duration.
- Pricing variation helps distinguish these channels.

Interpretation: This distinction is the core empirical contribution. Down payments work both because they reveal borrower type and because they change repayment incentives.

5.4 What the design can and cannot identify

The paper is strongest for counterfactuals that change down payment requirements and car markups near the observed support. It is less suited for welfare analysis or large counterfactuals involving consumer sophistication, rational expectations, or major changes in car choice.

Interpretation: The authors trade off a fully structural model of intertemporal utility for a tractable model that fits the observed pricing margins and is directly useful for profit and contract-design analysis.

6 Tables and figures

6.1 Table I: Summary statistics

Read:

- Applicants have low income and limited access to formal credit; only a minority have strong credit histories.
- The close rate is roughly one-third of applicants.
- Among buyers, the average down payment is around \$1000 and the average loan amount is close to \$11,000.
- Among uncensored loans, default is common and recovery values are low relative to car cost.

Economic message: The table establishes why contract design matters. Small changes in liquidity requirements can screen borrowers, while default and recovery outcomes determine the lender's realized profits.

TABLE I
SUMMARY STATISTICS^a

	Obs.	Mean	Std. Dev.	5%	95%
<i>Applicant Characteristics</i>					
Age	<i>N</i>	32.8	10.7	19	53
Monthly income (\$U.S.)	<i>N</i>	2414	1074	1299	4500
Home owner	<i>N</i>	0.15	—	—	—
Live with parents	<i>N</i>	0.18	—	—	—
Bank account	<i>N</i>	0.72	—	—	—
Risk category			—	—	—
Low	<i>N</i>	0.27	—	—	—
Medium	<i>N</i>	0.45	—	—	—
High	<i>N</i>	0.29	—	—	—
Car purchased	<i>N</i>	0.34	—	—	—
<i>Buyer Characteristics</i>					
Age	0.34 <i>N</i>	34.7	10.8	20	55
Monthly income	0.34 <i>N</i>	2557	1089	1385	4677
Home owner	0.34 <i>N</i>	0.17	—	—	—
Live with parents	0.34 <i>N</i>	0.16	—	—	—
Bank account	0.34 <i>N</i>	0.76	—	—	—
Risk category			—	—	—
Low	0.34 <i>N</i>	0.35	—	—	—
Medium	0.34 <i>N</i>	0.47	—	—	—
High	0.34 <i>N</i>	0.17	—	—	—
<i>Car Characteristics</i>					
Acquisition cost (\$U.S.)	0.34 <i>N</i>	5090	1329	3140	7075
Total cost (\$U.S.)	0.34 <i>N</i>	6096	1372	4096	8212
Car age (years)	0.34 <i>N</i>	4.3	1.9	2.0	8.0
Odometer (miles)	0.34 <i>N</i>	68,775	22,091	31,179	102,299
Time on lot (days)	0.34 <i>N</i>	33	44	1	122
Car price (\$U.S.)	0.34 <i>N</i>	10,777	1797	8095	13,595
<i>Transaction Characteristics</i>					
Min. down payment (applicants)	<i>N</i>	648	276	400	1200
Min. down payment (buyers)	0.34 <i>N</i>	750	335	400	1400
Interest rate (APR)	0.34 <i>N</i>	26.2	4.4	17.7	29.9
Loan term (months)	0.34 <i>N</i>	40.5	3.7	35.0	45.0
Down payment	0.34 <i>N</i>	942	599	400	2000
Loan amount	0.34 <i>N</i>	10,740	1801	7982	13,559
Monthly payment	0.34 <i>N</i>	395	49	314	471
<i>Loan Outcomes (uncensored sales only)</i>					
Default	0.13 <i>N</i>	0.61	—	—	—
Fraction of payments made	0.13 <i>N</i>	0.58	0.38	0.04	1.00
Loan payments	0.13 <i>N</i>	7972	5635	491	16,587
Nonzero recovery (all defaults)	0.08 <i>N</i>	0.78	—	—	—
PV of recovery (all recoveries)	0.06 <i>N</i>	1579	1328	231	4075
Gross operating revenue	0.13 <i>N</i>	9614	5192	2169	17,501
Net operating revenue	0.13 <i>N</i>	3333	5020	−3906	10,284

^aLoan payments, gross operating revenues, and net operating revenues are in present value (PV) terms. Gross operating revenue is equal to the sum of the down payment and the present value of the loan payments and recovery amount, assuming an internal firm discount rate of 10 percent. Net operating revenue is equal to gross operating revenue minus total car cost (including sales tax). To preserve the confidentiality of the company that provided the data, we do not report the exact number of applications.

Table I: Summary statistics.

6.2 Figures 1 and 2: Profit distribution and repayment risk

Read:

- Figure 1 shows a bimodal return distribution: paid loans are profitable, while defaulted loans generate large losses.
- Figure 2 shows that repayment probability declines with borrower risk and is higher for borrowers who voluntarily pay more than the minimum down payment.
- The gap between minimum-down and more-than-minimum borrowers suggests that borrower financing choices reveal private information.

Economic message: The lender's problem is not simply to maximize sales. The identity and leverage of the borrower determine whether a sale is profitable.

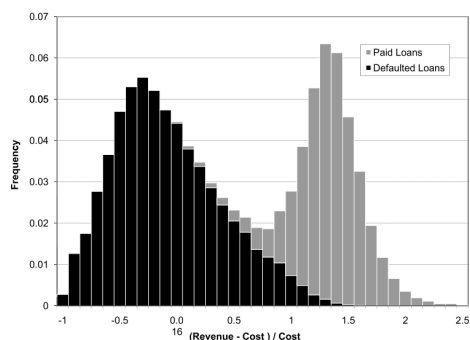


FIGURE 1.—Rate of return histogram. A unit of observation is a loan, and the figure is based on uncensored loans only. Revenue is calculated as the sum of the down payment, the value of the loan payments, and the present value of the recovery amount, assuming a firm discount rate of 10 percent. Cost includes the purchase cost at the auction and financing cost.

Figure 1: Rate-of-return histogram.

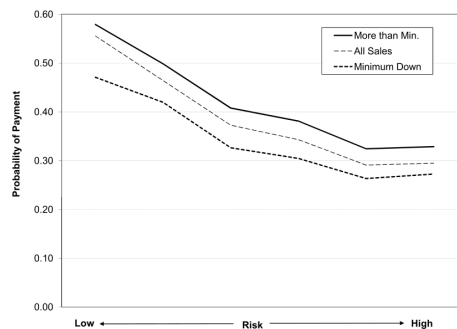


FIGURE 2.—Probability of payment by risk type and down payment. The figure is based on data from uncensored loans only. The horizontal axis represents a (discrete) measure of buy riskiness based on the internal company credit scoring. The vertical axis represents the probability that buyers in that risk category repay the loan in full. The “All Sales” line (long-dashed) shows the relationship between risk category and the probability of payment for all buyers. The “More than Min.” line (solid) shows the relationship for buyers who put down more than the required minimum down payment. The “Minimum Down” line (short-dashed) shows this relationship for buyers who put down exactly the required minimum down payment.

Figure 2: Probability of payment by risk type and down payment.

6.3 Figures 3 and 4: Pricing variation and model intuition

Read:

- Figure 3 documents discrete changes in minimum down payment requirements across risk categories and over time.
- Figure 4 illustrates the model’s logic: some consumers do not buy, some bind at the minimum down payment, and some voluntarily make larger down payments.
- Raising the minimum down payment screens out low-liquidity marginal borrowers; raising price mostly increases loan size.

Economic message: The paper’s identification comes from contract changes that shift different margins of borrower behavior. The model predicts that down payments and price markups should not have symmetric effects.

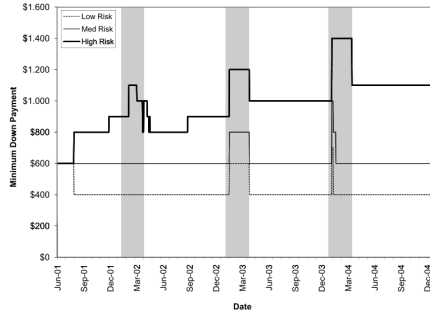


FIGURE 3.—Minimum down payments over time. Shaded areas represent “tax season” (mid-February through end of March) during which volume is higher and down payment requirements increase. Each line represents the time series of the required down payment for a *representative* high-risk, medium-risk, and low-risk applicant, based on the firm’s credit scoring. Other credit categories (not shown in the figure) follow a similar time series.

Figure 3: Minimum down payments over time.

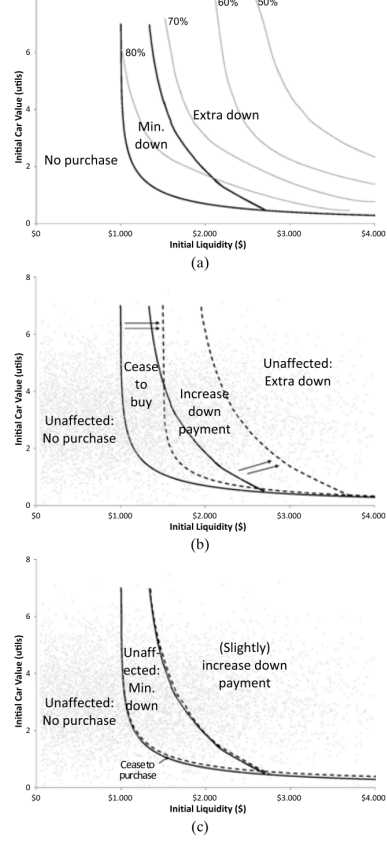


FIGURE 4.—Model illustration. Panel (a) shows consumer purchasing and down payment behavior. Panel (b) shows the distribution of consumers as a scatterplot, and the effect of changes in the minimum down payment. Panel (c) shows the effect of an increase in the car price.

Figure 4: Model illustration.

6.4 Table II and Figure 5: Model fit

Read:

- Table II compares raw data moments to moments predicted by the model.
- The model fits close rates, minimum down payment probabilities, average loan sizes, payment probabilities, and fraction of payments made reasonably well.
- Figure 5 shows that the model also matches the distribution of down payments and the distribution of repayment lengths.

Economic message: The estimated demand system captures the main margins needed for pricing: purchase, loan size, and repayment.

TABLE II
MODEL FIT^a

	Raw Data	Demand Model
<i>Close Rate</i>		
All applicants	0.343	0.339
Low risk	0.451	0.440
Medium risk	0.398	0.391
High risk	0.249	0.250
<i>Probability of Minimum Down Payment</i>		
All buyers	0.431	0.461
Low risk	0.234	0.335
Medium risk	0.428	0.463
High risk	0.570	0.566
<i>Average Loan Size</i>		
All buyers	\$10,709	\$10,674
Low risk	\$11,047	\$10,837
Medium risk	\$10,660	\$10,636
High risk	\$9992	\$10,093
<i>Probability of Payment</i>		
All buyers	0.390	0.405
Low risk	0.559	0.551
Medium risk	0.363	0.378
High risk	0.289	0.330
<i>Fraction of Payments Made</i>		
All buyers	0.594	0.620
Low risk	0.715	0.740
Medium risk	0.576	0.599
High risk	0.521	0.554

^aRaw data moments are computed directly from the estimation sample (see notes to Table III for details). Demand model moments are computed based on the econometric model described in Section 4 and the parameter estimates presented in Table III. The close rate, or probability of sale, is computed using data on all applicants. The probability of making minimum down payment and average loan size are conditional on sale. The probability of payment and the fraction of payments made are computed using data on uncensored loans only.

Table II: Model fit.

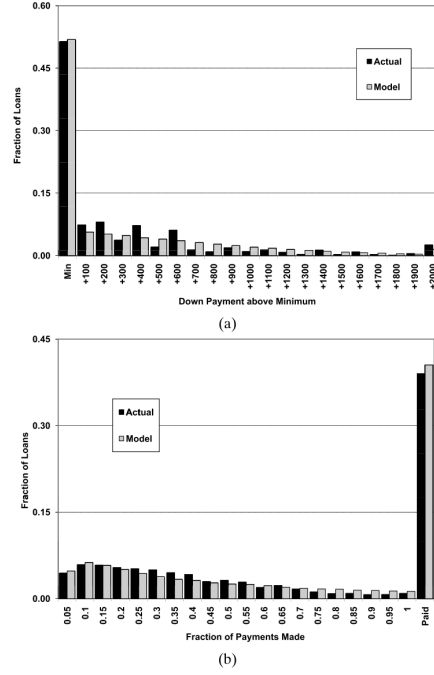


FIGURE 5.—Model fit. Panel (a) shows the model fit for down payment decisions. The last bin includes all down payments of at least \$2000 above the minimum down. Panel (b) shows the model fit for the fraction of loan payments made. Note that the model slightly overstates the action of consumers who default just prior to paying off their loan.

Figure 5: Model fit.

6.5 Table III: Demand estimates

Read:

- A \$100 increase in the minimum down payment lowers sale probability by about 2.3 percentage points.
- A \$100 increase in negotiated price has little effect on sale probability and raises desired down payment by only about \$11.
- A \$1000 larger loan reduces the fraction of payments made by about 32 percent.
- Credit categories strongly predict repayment: low-risk borrowers make substantially more payments than high-risk borrowers.
- The covariance estimates show that unobserved purchase and down-payment choices contain information about repayment risk.

Economic message: Consumers are cash constrained and debt sensitive. They react strongly to cash due today, but price increases are largely converted into debt, which then raises default risk.

TABLE III
DEMAND ESTIMATES^a

	Probability of Sale		Down Payment		Payments Made	
Dependent Variable	Marg. Eff.	Std. Err.	Coeff.	Std. Err.	Coeff.	Std. Err.
<i>Offer Variables</i>						
Minimum down (\$1000s)	-0.231	(0.011)	—	—	—	—
Negotiated price (\$1000s)	-0.025	(0.006)	0.106	(0.086)	—	—
Maximum interest rate (%)	0.002	(0.002)	0.024	(0.004)	-0.030	(0.006)
Term (years)	-0.060	(0.011)	-0.375	(0.100)	0.070	(0.101)
Loan amount (\$1000s)	—	—	—	—	-0.386	(0.041)
<i>Car Characteristics</i>						
Car cost (\$1000s)	0.011	(0.007)	0.162	(0.087)	0.389	(0.043)
Premium (Cost > \$7500)	0.013	(0.009)	0.458	(0.036)	0.059	(0.070)
Car age (years)	0.001	(0.001)	0.000	(0.005)	-0.049	(0.009)
Odometer (10,000s)	-0.003	(0.001)	0.000	(0.004)	0.005	(0.008)
Lot age (months)	0.001	(0.002)	-0.041	(0.017)	-0.079	(0.011)
<i>Individual Characteristics</i>						
Income (\$1000s/month)	0.021	(0.002)	-0.002	(0.009)	0.078	(0.014)
Age	0.007	(0.002)	-0.015	(0.004)	0.014	(0.007)
Age squared	-6E-05	(2E-05)	2E-04	(5E-05)	-1E-04	(8E-05)
Bank account	0.034	(0.005)	0.036	(0.017)	0.214	(0.031)
House owner	-0.032	(0.006)	0.006	(0.019)	0.008	(0.042)
Lives with parents	0.003	(0.007)	0.043	(0.019)	-0.111	(0.036)
<i>Credit Grade Fixed Effects</i>						
Representative low-risk grade	-0.012	(0.017)	0.349	(0.060)	0.869	(0.101)
Representative medium-risk grade	0.004	(0.014)	0.260	(0.056)	0.316	(0.081)
Representative high-risk grade	Omitted		Omitted		Omitted	
<i>Seasonal Effects</i>						
Tax season	0.137	(0.010)	0.545	(0.027)	0.023	(0.065)
<i>Other Fixed Effects</i>						
	Year, Month, City, Credit Grade					
<i>Covariance Matrix</i>						
	ε_Q		ε_D		ε_S	ε_P
ε_Q (purchase equation)	1.000	—	0.062 (0.007)	0.003 (0.014)	-0.018 (0.006)	
ε_D (down payment equation)	0.062 (0.007)		0.557 (0.027)	0.022 (0.014)	-0.024 (0.058)	
ε_S (repayment equation)	0.003 (0.014)		0.022 (0.014)	2.334 (0.047)	0.078 (0.015)	
ε_P (car price equation)	-0.018 (0.006)		-0.024 (0.058)	0.078 (0.015)	0.283 (0.020)	

^aAll estimates are based on the demand model described in Section 4. The sample for the purchase and down payment equations is a random sample of all applicants; the sample size is $0.1N$, where $N \gg 50,000$ (see Table I). The sample for the fraction of payments made equation is all sales; sample size is $0.034N$. Reported estimates in the second column show the marginal effects of a 1 unit change in each of the explanatory variables on the probability of sale. For dummy variables, this is computed by taking the difference between the probability of sale when the variable is equal to 1 and the probability when the variable is equal to 0 (holding other variables fixed). For continuous variables, this is computed by taking a numerical derivative of the probability of sale with respect to the continuous variable. Estimates in the fourth column show the effects of a 1 unit change in each explanatory variable on the desired down payment (in \$1000s). For instance, a \$1000 increase in price raises the desired down payment of the average applicant by \$106. Estimates in the sixth column show the effects of a 1 unit change in each explanatory variable on the log of fraction of payments made. For example, a \$1000 increase in loan amount decreases the fraction of payments made by $1 - \exp(-0.386)$, or 32 percent. Standard errors are based on 60 bootstrap samples.

Table III: Demand estimates.

6.6 Tables IV and V: Recovery and implied indirect costs

Read:

- Table IV shows that recoveries depend heavily on car characteristics and the number of months paid before default.
- Table V estimates the indirect cost needed to rationalize observed down payment policies.
- The baseline implied indirect cost is about \$2481 per loan, with estimates varying by risk category and under alternative learning assumptions.

Economic message: Observed down payment requirements look conservative relative to direct car costs alone. The paper interprets this as consistent with servicing costs, aggregate portfolio risk, securitization frictions, or other indirect costs of lending.

Dependent Variable	Nonzero Recovery Ind. Probit		Recovery Amt. (\$1000s) OLS	
	dF/dx	Std. Err.	Coeff.	Std. Err.
Months paid	-0.006	(0.0002)	-0.045	(0.001)
<i>Car Characteristics</i>				
Car cost (\$1000s)	0.027	(0.002)	0.526	(0.006)
Premium (cost > \$7500)	-0.097	(0.008)	0.079	(0.024)
Car age (years)	-0.008	(0.001)	-0.068	(0.004)
Odometer (10,000s)	0.010	(0.001)	-0.029	(0.003)
Lot age (months)	-0.004	(0.001)	-0.131	(0.004)
<i>Individual Characteristics</i>				
Income (\$1000s/month)	0.000	(0.002)	0.048	(0.006)
Bank account	0.001	(0.003)	0.078	(0.012)
House owner	0.001	(0.004)	0.118	(0.015)
Other fixed effects	Year, Month, City		Year, Month, City	

^aThe sample for the probit equation is all defaults; sample size is 0.18N, where $N \gg 50,000$ (see Table I). Reported coefficients show the marginal effect of a 1 unit change in the explanatory variable on the probability of making a positive recovery. The sample for the ordinary least squares (OLS) equation is all recoveries; sample size is 0.14N.

Table IV: Recovery parameter estimates.

	Cost Estimate	Standard Error
Uniform	2481	(68)
Credit Category		
Low risk	3265	(111)
Medium risk	2373	(74)
High risk	1932	(88)
Seasonal		
Non-tax season	2498	(70)
Tax season	2285	(151)
Learning Assumptions		
Assumption (i)	1800	(332)
Assumption (ii)	2200	(240)
Assumption (iii)	2800	(222)
Assumption (iv)	2600	(207)

^aAll results are estimates of the firm's unobserved indirect cost of sale. The top six rows are based on the first-order condition for pricing described in Section 5. Estimates assume an internal firm discount rate of 10 percent. Standard errors are based on 60 bootstrap samples. The four estimates, which are based on "learning" inequalities, reflect the following conditions: (i) period $t+1$ pricing dominates period t pricing in period $t+1$; (ii) period $t+1$ pricing dominates period t pricing in period $t+1$ (weighted); (iii) period $t+1$ pricing dominates period t pricing in period t (weighted); (iv) period $t+1$ pricing dominates period t pricing in period t (weighted). In each case, the estimates are computed by minimizing the sum of squared violations of the inequality dictated by each condition. For indirect cost estimates (ii) and (iv), violations are weighted by the number of applicants in a given period. For example, to compute learning model (i), we compute expected profits per applicant in each pricing period using observed prices in that period and observed prices from the previous period. If the updated prices result in higher profits than the previous prices, then there is no violation of the learning inequality. If updated prices performed worse than previous prices, then we calculate the loss due to the pricing change. The sum of squared losses across all pricing changes is then minimized by searching over a grid of indirect costs, and the minimizing indirect cost is our estimate.

Table V: Estimates of indirect costs.

6.7 Figures 6 and 7: Contract-pricing trade-offs

Read:

- Figure 6 shows that increasing minimum down payments lowers sales but also lowers default probability.
- The profit-maximizing down payment is lower for low-risk applicants and higher for high-risk applicants.
- Figure 7 shows that increasing price-cost margins has little effect on sales, but substantially increases default risk by increasing loan balances.
- Optimal margins are higher for low-risk applicants because their repayment is less sensitive to additional debt.

Economic message: Down payment requirements mainly trade off volume against borrower quality. Markups mainly trade off payment size against repayment probability. These are distinct contract-pricing margins.

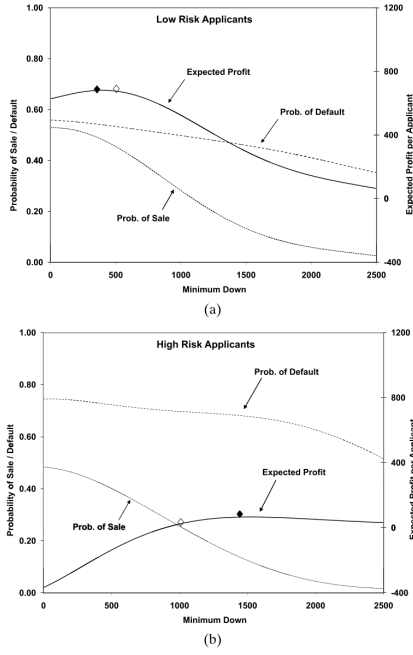


FIGURE 6.—Effect of minimum down payment changes. The figure is based on model estimates for all low-risk (a) and high-risk (b) applicants. The horizontal axis represents the required minimum down payment applied to applicants in each risk category. The left-hand y axis represents the probability of sale (for applicants) and the probability of default (for buyers). The right-hand y axis represents expected profit per applicant, calculated as the probability of sale times net operating revenue per sale, where net operating revenue is equal to the sum of the down payment and the present value of loan payments and recoveries, minus total cost (observed cost and unobserved cost). The unobserved cost is estimated as described in Section 5. Open diamonds show observed average minimum down payments for each credit category. Solid diamonds show optimal minimum down payments based on the model estimates.

Figure 6: Effect of minimum down payment changes.

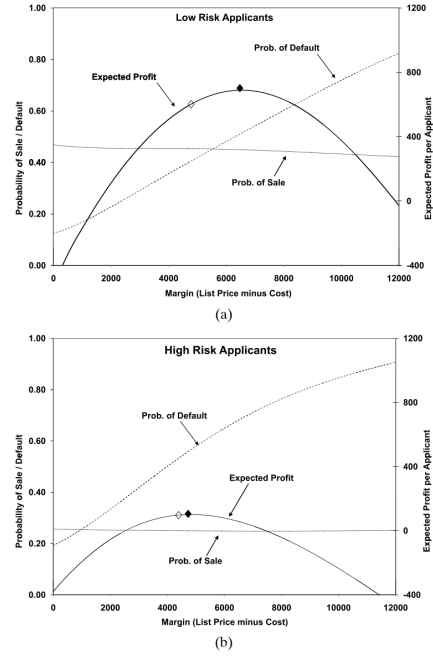


FIGURE 7.—Effect of price-cost margin changes. The figure is based on model estimates for all low-risk (a) and high-risk (b) applicants. The horizontal axis represents the target margin (list price minus cost) applied to applicants in each risk category. The left-hand y axis represents the probability of sale (for applicants) and the probability of default (for buyers). The right-hand y axis represents expected profit per applicant, calculated as the probability of sale times net operating revenue per sale, where net operating revenue is equal to the sum of the down payment and the present value of loan payments and recoveries, minus total cost (observed cost and unobserved cost). The unobserved cost is estimated as described in Section 5. Open diamonds show the observed average target margin for each credit category. Solid diamonds show the optimal target margin based on the model estimates.

Figure 7: Effect of price-cost margin changes.

6.8 Table VI and Figure 8: Value of credit scoring

Read:

- Table VI compares observed pricing, optimal credit-based pricing, optimal uniform pricing, and pricing with perfect information.
- Optimal credit-based pricing assigns lower required down payments to safer borrowers and higher required down payments to riskier borrowers.
- Relative to optimal uniform pricing, risk-based pricing raises expected profit; perfect information would raise profits much further.
- Figure 8 illustrates how risk-based down payments reduce high-risk loan originations, force higher-risk borrowers to make larger down payments, and lower default rates.

Economic message: Credit scoring is valuable because it allows the lender to screen and size loans by risk. But borrowers still have private information that credit scores do not capture.

TABLE VI
VALUE OF CREDIT SCORING^a

	Low Risk	Med. Risk	High Risk	All Applicants
<i>Minimum Down Payment</i>				
Observed pricing	\$400	\$600	\$1000	—
Optimal credit-based pricing	\$0	\$700	\$1550	—
Optimal uniform pricing	\$800	\$800	\$800	\$800
Pricing with perfect information	—	—	—	—
<i>Close Rate</i>				
Observed pricing	0.451	0.398	0.249	0.343
Optimal credit-based pricing	0.505	0.355	0.119	0.305
Optimal uniform pricing	0.345	0.355	0.312	0.339
Pricing with perfect information	0.504	0.443	0.266	0.394
<i>Profit Conditional on Sale</i>				
Observed pricing	\$1967	\$797	\$106	\$996
Optimal credit-based pricing	\$1840	\$944	\$679	\$1201
Optimal uniform pricing	\$2137	\$944	-\$34	\$883
Pricing with perfect information	\$2292	\$1333	\$918	\$1490
<i>Expected Profit per Applicant</i>				
Observed pricing	\$886	\$317	\$26	\$341
Optimal credit-based pricing	\$930	\$335	\$81	\$366
Optimal uniform pricing	\$736	\$335	-\$11	\$299
Pricing with perfect information	\$1156	\$591	\$244	\$587

^a All results are based on model estimates. Close rate is the probability that an applicant purchases a car. Profit conditional on sale is defined as net operating revenue (the sum of down payment and the present value of loan payments and recoveries, minus vehicle cost) minus our estimate of indirect cost in the top row of Table V. Expected profit per applicant is equal to the close rate times profit conditional on sale. Each counterfactual represents a different minimum down payment policy. List prices are held fixed at observed values in all counterfactuals. Observed pricing describes outcomes based on the company's observed minimum down payments, which vary both over time and across credit categories. Optimal credit-based pricing describes a counterfactual in which minimum down payments vary by credit category so as to maximize expected profit per applicant in each observed pricing period. Optimal uniform pricing describes a counterfactual in which a single minimum down payment, which is constant across credit categories, is chosen to maximize expected profit per applicant in each pricing period. Pricing with perfect information describes a counterfactual in which the firm can observe the borrower's unobservables at the time of purchase and sets a minimum down payment for each applicant equal to the maximum amount that the applicant is able to put down, and sells only to applicants with positive expected profits at this minimum down payment.

Table VI: Value of credit scoring.

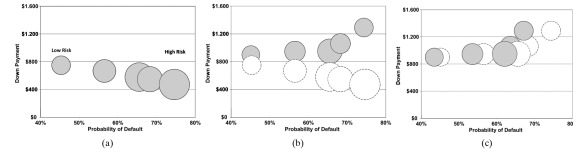


FIGURE 8.—The relationship between down payments and default rates under different conditions. The figure shows model-estimated average down payments and default rates for borrowers in different risk categories under two different pricing regimes: one with no minimum down payment and one with observed minimum down payments that vary by risk category. Each bubble represents an observable risk category defined by the company's proprietary credit score. The size of each bubble represents the estimated loan volume for a given risk category. Estimated average down payments, default rates, and loan volumes are calculated for all applicants based on the demand model estimated in the paper. Panel (a) shows estimated average down payments, default rates, and loan volumes assuming that all applicants face observed car prices and minimum down payment. Panel (b) shows estimated average down payments assuming applicants in each risk category face their observed average minimum down payment, keeping the default rates as in panel (a). Panel (c) adjusts panel (b) to include estimated default rates based on observed average minimum down payments (the average down payments and bubble sizes are the same as in panel (b)).

Figure 8: Down payments and default rates under different pricing regimes.

6.9 Table VII: Credit scoring as a barrier to entry

Read:

- Table VII studies monopoly and duopoly scenarios in which the incumbent and entrant may or may not price by risk category.
- An incumbent with risk-based pricing earns higher per-applicant profits than one forced to price uniformly.
- Proprietary scoring can reduce entrant profits by leaving the entrant with an adversely selected borrower pool.

Economic message: Credit scoring is not only a pricing technology; it can also be a strategic asset that changes competitive selection in credit markets.

TABLE VII
CREDIT SCORING AS A BARRIER TO ENTRY^a

	(Incumbent Profit per Applicant, Entrant Profit per Applicant)	
	Incumbent Prices Uniformly	Incumbent Prices by Risk Category
No entrant (monopoly)	(\$299, \$0)	(\$366, \$0)
Entrant prices uniformly	(\$132, \$132)	(\$196, \$93)
Entrant prices by grade	(\$93, \$196)	(\$166, \$166)
	Equilibrium Minimum Down Payments	
	Uniform	By Risk Category (Low, Med., High)
Monopoly	\$750	\$50, \$750, \$1450
Versus uniform	\$450	\$50, \$400, \$1000
Versus risk-based	\$750	\$0, \$400, \$1050

^aAll results are based on model estimates. Each cell in the first panel of the table presents the expected profits per applicant for an incumbent lender (first) and the profits per applicant for a potential entrant (second), calculated at the equilibrium minimum down payments shown in the corresponding cell of the second panel. The top row presents the case of no entrant, which is also presented in Table VI. The second row presents the case where an entrant prices uniformly (i.e., sets one minimum down payment for all risk categories), and the third row represents the case where an entrant prices by risk category. Each column represents the pricing strategy of the incumbent firm. In each scenario, we find the Nash equilibrium of the duopoly game in which each firm simultaneously either sets uniform or risk-based minimum down payments. We assume that applicants who choose to put down more than either firm's minimum down payment are randomly split between the two firms, and other applicants choose the lender with the lowest minimum down payment. We also assume that car prices remain the same in all scenarios. Expected profits conditional on sale are then calculated using the estimated repayment equation (Table III, sixth column) and estimated indirect costs (Table V).

Table VII: Credit scoring as a barrier to entry.

7 Short summary

- The paper studies contract pricing in subprime auto lending using detailed transaction-level data.
- The model jointly estimates purchase, down payment, and repayment decisions.
- The key contract terms are minimum down payment requirements and car price markups.
- Down payment requirements strongly reduce purchase probability, screen out riskier marginal borrowers, and reduce loan sizes.
- Car price increases have little effect on sales but increase loan sizes and therefore default risk.
- Borrowers who voluntarily make larger down payments are better risks, even conditional on observed credit category.
- Larger loans causally worsen repayment; a \$1000 larger loan reduces the fraction of payments made by about 32 percent.
- Observed lending appears conservative relative to direct car costs, implying substantial indirect costs of lending.
- Risk-based financing raises profits relative to uniform down payment requirements.
- Perfect information would generate much larger profits, suggesting that consumer private information remains important.

8 Conclusion

8.1 Core conclusion

Einav, Jenkins, and Levin show that credit-contract pricing must be analyzed as a multi-dimensional problem. In subprime auto lending, down payment requirements, markups, interest rates, and loan sizes affect not only demand but also borrower selection and repayment behavior. Down payment requirements are particularly powerful because they operate through current liquidity and leverage.

8.2 Economic intuition

Subprime borrowers have limited cash but strong demand for transportation. A higher down payment directly raises the amount of cash needed today, so it sharply reduces purchases and screens out borrowers with low liquidity. A higher car price, however, can be financed, so many borrowers still buy but take larger loans. Larger loans increase monthly payment burdens and make default more likely. This difference explains why down payments and markups have different optimal pricing implications.

8.3 Incremental contribution revisited

The paper's central contribution is to connect asymmetric information, borrower liquidity, and imperfect-competition pricing in one empirical framework. Rather than simply documenting adverse selection, it asks how the lender should set contract terms when borrowers differ in observable and unobservable risk and when contract terms change both who borrows and how much they repay.

8.4 Implications for consumer credit markets

The results suggest that relaxed down payment requirements can generate rapid credit expansion while worsening borrower quality and increasing leverage. This is relevant beyond auto lending, including mortgage and other household-credit markets. The paper also implies that policy or empirical analysis that focuses only on interest rates misses important contract dimensions.

8.5 Implications for firms and competition

For lenders, credit scoring is valuable because it permits more precise risk-based contract terms. It also has strategic value: a lender with better information can select profitable borrowers and leave competitors with a worse pool. Thus, information technology can affect both pricing and market structure.

8.6 Limitations

The analysis is based on one lender in a specific subprime auto market. The model is designed for pricing counterfactuals around observed contract terms and is less suited for broad welfare analysis or large counterfactuals involving consumer sophistication, learning, or changes in car choice. The estimates also rely on parametric assumptions about unobserved heterogeneity.

8.7 Takeaway

The main takeaway is that in high-risk credit markets, the right object is not simply demand elasticity but contract-profit elasticity: how a contract term changes volume, borrower quality, loan size, repayment, recovery, and profit. Down payments are a central screening and incentive device, and credit scoring is valuable because it helps allocate this device across borrowers with different risk.